ESPaperPlay / DEVELOPMENT BOARD r5

ESP32-S3-WROOM-1-N16R8 / GDEY075T7-T01 / protected 1-cell battery

Ten functional sheets. Schematic review stage; no PCB or manufacturing release.

POWER_INPUT_CHARGER

Input / charger / battery

文件: POWER_INPUT_CHARGER.kicad_sch

POWER_3V3_CONTROL

Regulator / IO expander / gauge

文件: POWER_3V3_CONTROL.kicad_sch

MCU_CORE

N16R8 / reset / user keys

文件: MCU_CORE.kicad_sch

USB_INPUT

USB-C / protection / source select

文件: USB_INPUT.kicad_sch

POWER_DOMAINS

Five switched power domains

文件: POWER_DOMAINS.kicad_sch

TOUCH_ISOLATION

GT911 four-signal isolation

文件: TOUCH_ISOLATION.kicad_sch

EPD_DISPLAY

UC8179 / boost / 24-pin FPC

文件: EPD_DISPLAY.kicad_sch

STORAGE_SD

microSD / bus isolation / detect

文件: STORAGE_SD.kicad_sch

AUDIO

I2S amplifier / microphone

文件: AUDIO.kicad_sch

CHARGE_TEMP

NTC window / hardware TS pause

文件: CHARGE_TEMP.kicad_sch

ASSEMBLY / REVIEW BOUNDARIES

0.70A nominal; external NTC window uses an assumed Honeywell curve.
R108 stays DNP; J105 open. No charging or manufacturing release.
J502 and J202 each use ONE shunt. Actual pack R-T / thermal tests pending.
Unchanged functional sheets retain their r4 title blocks and review history.

hardware/devboard-r0/README.md / docs/v2/hardware/power-design-spec.md

DESIGN REVIEW ONLY — NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

ESPaperPlay / xiaobaoaaaaaa

Sheet: /
File: ESPaperPlay_Devboard.kicad_sch

Title: ESPaperPlay DEVELOPMENT BOARD — PERIPHERAL DRAFT

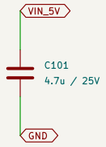
Size: A3	Date: 2026-11-04	**Rev: r5 DRAFT**
KiCad E.D.A. 10.0.7		Id: 1/11

1 POWER PATH AND PROTECTED BATTERY INTERFACE

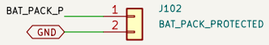
5 V only. USB_INPUT selects protected USB or J101 BENCH_5V into VIN_5V.



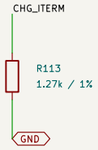
Known, current-limited 5 V bench source.
Select at J502; never bridge both sources.







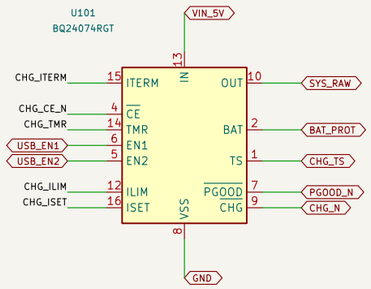




755570 / 4000mAh; assumed NTC curve.
BAT_NTC feeds external window, not U101 TS.
Verify sample curve and harness polarity.



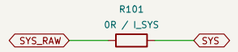




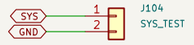
PWR_FLAG: normal RUN with R101/R203 populated.
ERC does not model removable links or input protection.



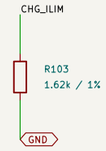


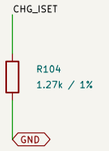


R101: measure total SYS load.
CE pull-up stays on SYS_RAW side.









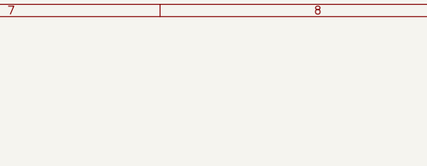


ISET 1.27k: ~701mA typical; external ITERM 1.27k.
TMR 68k: ~9.07h; DPPM / thermal may reduce current.
R108 stays DNP; verify temperature window and pack.

3 AON-SAFE VBUS SENSE



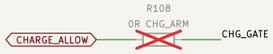


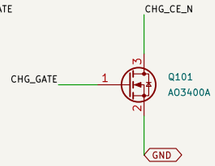


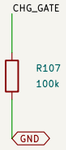


LOW = protected USB branch powered.
PGOOD reports selected input; PTC trip differs.

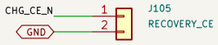
2 HARDWARE CHARGE INHIBIT / CURRENT MODE / STATUS







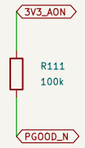








EN2 EN1: 00=100mA, 01=500mA,
10=RILIM, 11=USB suspend.
Do not infer source capacity from VBUS.





J105 normally OPEN. Recovery: MAIN_RUN OFF, AMP OFF, EN1/EN2=00.
J105 authorizes CE only; external window still inhibits via CHG_TS.
R108 DNP: firmware cannot enable charging in initial assembly.

DESIGN REVIEW ONLY — NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

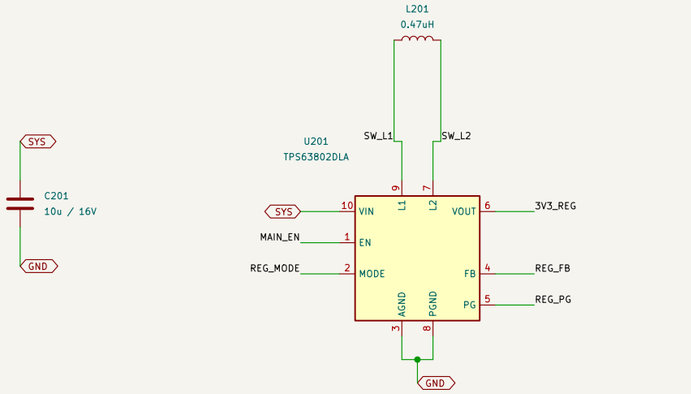
ESPaperPlay / xiaobaoaaaaaa

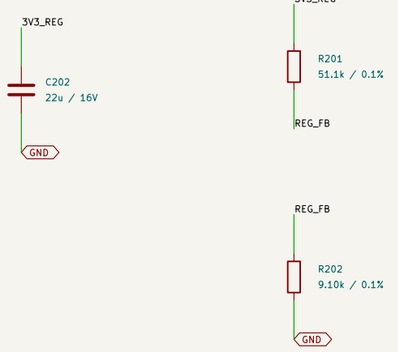
Sheet: /POWER_INPUT_CHARGER/
File: POWER_INPUT_CHARGER.kicad_sch

Title: POWER INPUT / BQ24074 CHARGER

Size: A3	Date: 2026-11-04	**Rev: r5 DRAFT**
KiCad E.D.A. 10.0.7		Id: 2/11

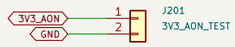
1 MAIN BUCK-BOOST / MEASUREMENT / RECOVERY

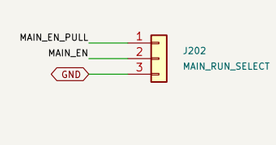




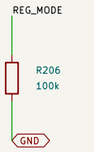


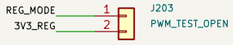
Never parallel an external 3.3 V source.
Open R203 before external injection.
C202 + R201 remain upstream of R203.



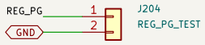








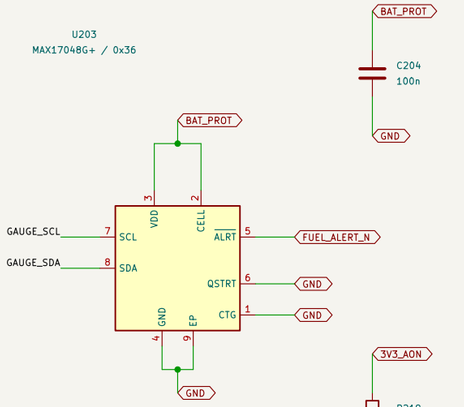




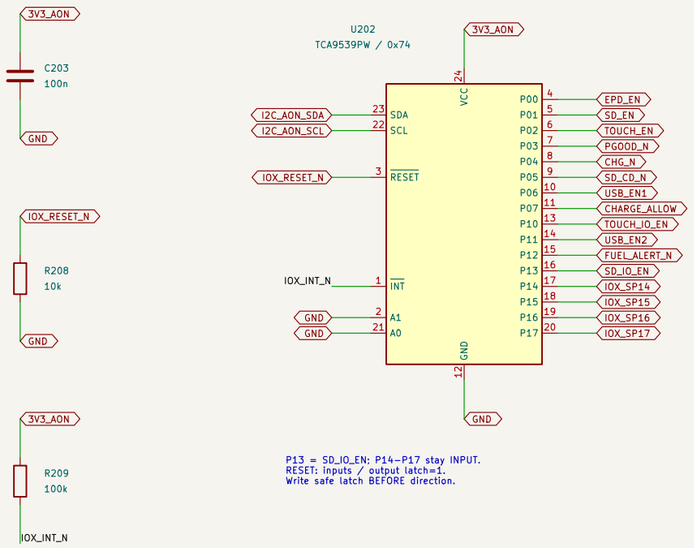
J202: 1-2 RUN; 2-3 OFF / recovery.
One shunt only. Open defaults OFF.

MODE low = automatic PFM/PWM.
U201 DLA vendor land assigned: assembly review open.
PG is a test signal, not a new MCU GPIO.

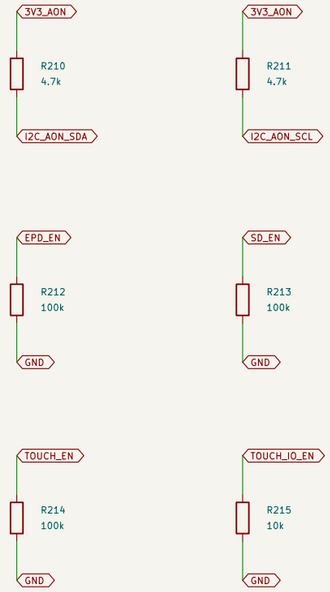
3 BATTERY GAUGE

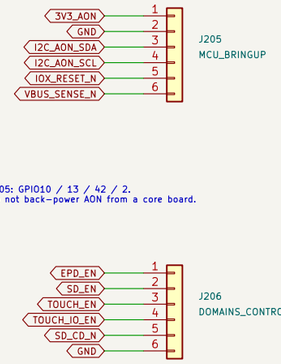


2 AON I2C CONTROL / RESET DEFAULTS



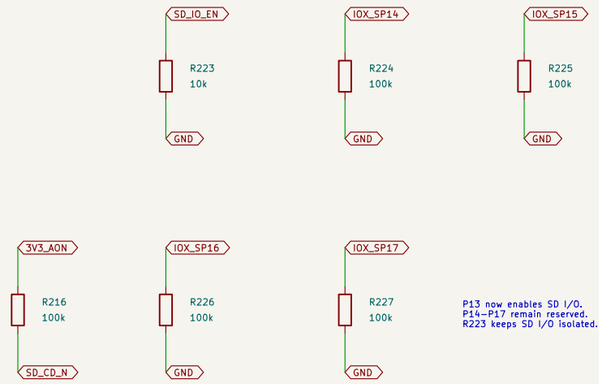
P13 = SD_IO_EN; P14-P17 stay INPUT.
RESET: inputs / output latch=1.
Write safe latch BEFORE direction.





J205: GPIO10 / 13 / 42 / 2.
Do not back-power AON from a core board.

4 SPARE PORT DEFAULTS



P13 now enables SD I/O.
P14-P17 remain reserved.
R223 keeps SD I/O isolated.

MCU, load switches and touch isolation are now connected across sheets.
Start I2C at 100kHz; edge timing / gauge power-off leakage to verify.

DESIGN REVIEW ONLY — NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

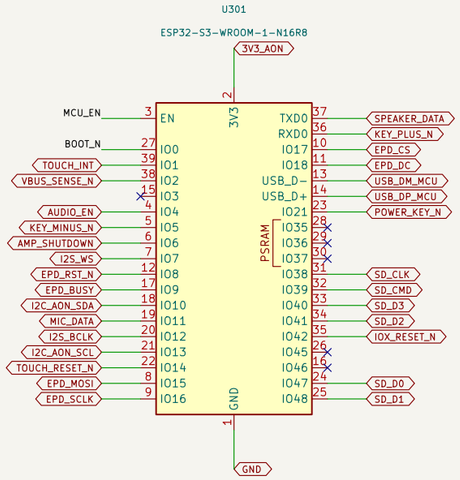
ESPaperPlay / xiaobaoaaaaaa

Sheet: /POWER_3V3_CONTROL/
File: POWER_3V3_CONTROL.kicad_sch

Title: 3.3 V REGULATOR / POWER CONTROL / GAUGE

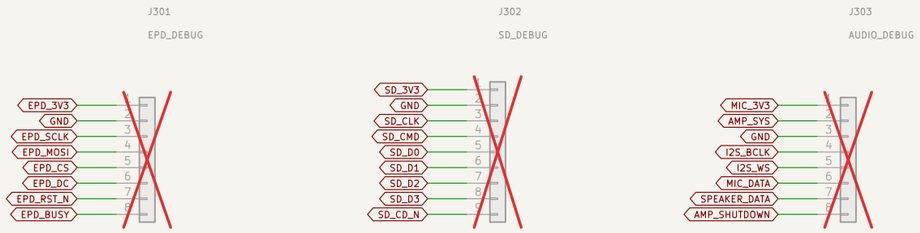
Size: A3	Date: 2026-11-02	**Rev: r4 DRAFT**
KiCad E.D.A. 10.0.7		Id: 3/11

1 MCU / N16R8 PHYSICAL PAD MAPPING



NC: GPIO3 / 45 / 46 (straps); GPIO35 / 36 / 37 (PSRAM).
Do not substitute WROOM-1-N16R16V: pad voltage differs.
Pads 24 / 25 = GPIO47 / 48, respectively.
Antenna at PCB edge; final keepout per module land pattern.

2 MCU-SIDE DEBUG HEADERS / ON-BOARD LOADS CONNECTED



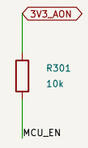
DNP DEBUG: high-impedance probes only; never attach competing peripheral modules.
Audio: MIC_3V3 and AMP_SYS are different supplies; check module voltage.
MCU-side signals: on-board buffers / switches are on each peripheral sheet.
Module pads 1 / 40 / 41 (EPAD) to GND: PCB antenna keepout required.

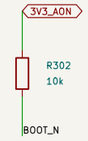
3 MAINTENANCE RESET / DOWNLOAD

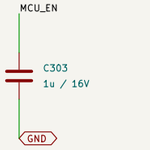
LOCAL BYPASS







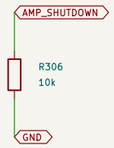




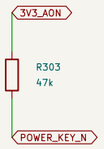




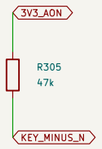
EN: 10k / 1uF initial RC; validate actual power ramp.
BOOT: pull-up only, no large capacitor.
GPIO43 / 44 are assigned: no always-connected UART0.
Native USB Serial/JTAG is the default recovery/debug path.



4 USER KEYS / SOFTWARE DEBOUNCE













IO21 is the normal sleep/wake key. EN is a separate service control.
I2C pull-ups and IOX_RESET_N pull-down live on POWER_3V3_CONTROL.
EPD / SD / touch / audio signals are global labels to their owning sheets.

DESIGN REVIEW ONLY — NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

ESPaperPlay / xiaobaoaaaaaa

Sheet: /MCU_CORE/
File: MCU_CORE.kicad_sch

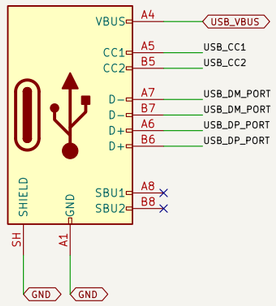
Title: ESP32-S3 N16R8 / RESET / USER KEYS

Size: A3	Date: 2026-11-02	**Rev: r4 DRAFT**
KiCad E.D.A. 10.0.7		Id: 4/11

1 USB-C DEVICE / NO PD / NO HOST POWER

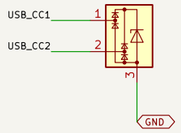
J501

USB-C USB2 16P



U502

TI TPD2E2U06DCKR



USB_CC1

R501

5.1k 1%

GND

USB_CC2

R502

5.1k 1%

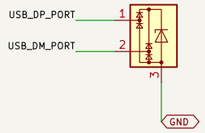
GND

CC1 / CC2 each have their own Rd.
A6+B6 = D+; A7+B7 = D-.
Shield bonded to local GND at connector; final EMC review required.

2 DATA ESD / MCU-SIDE MATCHING

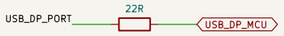
U501

TI TPD2E2U06DCKR



R503

22R





R504

22R



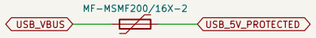


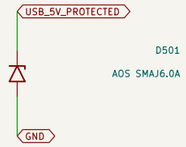
USB_DM_MCU -> GPIO19 / module pad13.
USB_DP_MCU -> GPIO20 / module pad14.
Target 90 ohm differential; short pair, continuous GND return.
USB default 100mA; firmware authorization precedes any 500mA mode.
CC resistors do not negotiate PD, or authorize 1.5A / 3A.
No ESD / USB compliance claim until board-level validation.

3 USB BRANCH PROTECTION / SINGLE-SHUNT SOURCE SELECT

F501

MF-MSMF200/16X-2

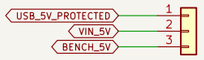






J502

ONE SHUNT: USB / BENCH



1-2 USB / 2-3 BENCH / OPEN = input disconnected.
VIN_5V goes to charger; protected USB is sensed before J502.
One shunt. Known regulated 5V, CURRENT-LIMITED bench input.

PTC / TVS protect the USB branch only: no sustained OVP or automatic source ORing.
TVS leakage / overshoot, fault current, VBUS inrush and cable drop require tests.
F501200/16X: 60C hold1.48A vs1.085A budget; fault/TVS and USB assembly tests remain.

DESIGN REVIEW ONLY - NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

ESPaperPlay / xiaobaoaaaaaa

Sheet: /USB_INPUT/
File: USB_INPUT.kicad_sch

Title: USB-C DEVICE / ESD / 5V SOURCE SELECTION

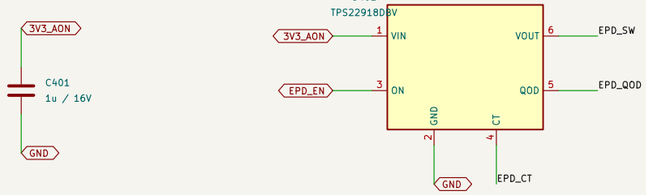
Size: A3	Date: 2026-11-02	**Rev: r4 DRAFT**
KiCad E.D.A. 10.0.7		Id: 5/11

FIVE TPS22918 DOMAINS / COMPONENT VALUES ARE UNVALIDATED CANDIDATES

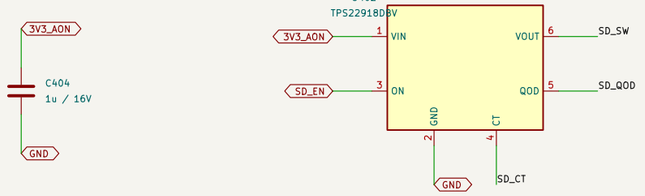
CT = 4.7nF / 50V COG 5%; RQOD = 330R + internal RPD. Links measure downstream load; switch IQ is upstream.

EPD/SD/TOUCH ON pulls: R212/R213/R214 on POWER_3V3_CONTROL. AUDIO_EN: R416 below. No reverse-current blocking.

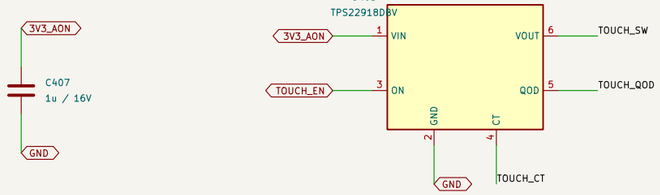
1 EPD



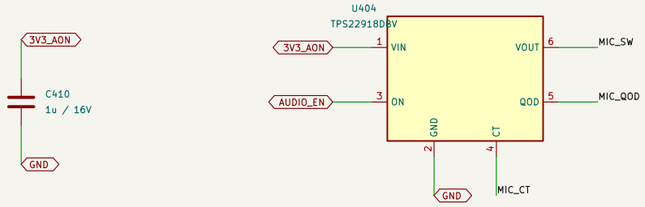
2 SD



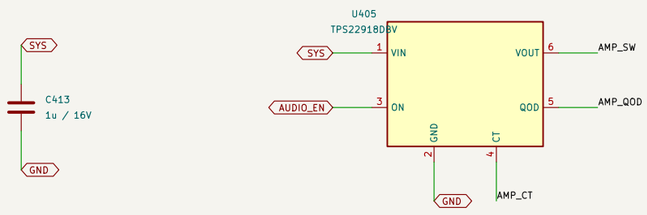
3 TOUCH



4 MIC



5 AMP





R416: both audio rails default OFF when GPIO4 is high impedance.
SYS may remain alive with AON=0; measure AMP leakage in this state.

DESIGN REVIEW ONLY — NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

ESPaperPlay / xiaobaoaaaaaa

Sheet: /POWER_DOMAINS/
File: POWER_DOMAINS.kicad_sch

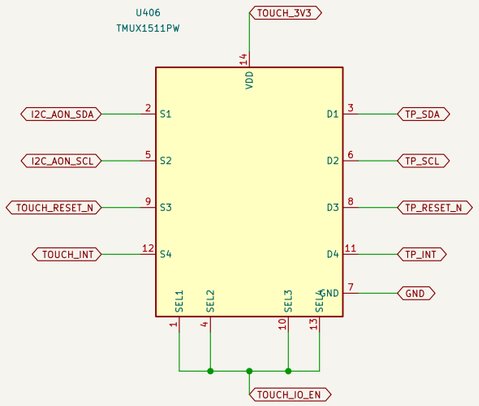
Title: PERIPHERAL POWER / FIVE SWITCHED DOMAINS

Size: A3	Date: 2026-11-02	**Rev: r4 DRAFT**
KiCad E.D.A. 10.0.7		Id: 6/11

1 TMUX1511PW: ALL FOUR SIGNALS ISOLATED / ACTIVE-HIGH SELECT

VDD, bypass, pull-ups and GT911 VCC all use TOUCH_3V3 AFTER R408 (I_TOUCH). Never power the switch from AON.

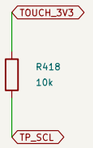




MCU / AON side
GPIO10 SDA; GPIO13 SCL
GPIO14 RESET; GPIO1 INT
R210/R211: 4.7k to AON

R215 = 10k to GND on control sheet.
All SELs LOW: isolated; HIGH: connected.
INT remains bidirectional for GT911 boot.

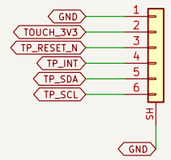












0.5mm FPC6 + GND retention pads.
Keep extension; candidate footprint set.
Check contact face / full-chain pin1.



PWR_FLAG: normal assembly with R408 fitted.
ERC does not model an open measurement link.

2 POWER / SIGNAL SEQUENCING AND MEASUREMENT

ON: keep TOUCH_IO_EN=0; power TOUCH; wait for valid rail; set MCU RESET/INT; connect TMUX; execute full GT911 reset.
OFF: stop access and touch wake; set TOUCH_IO_EN=0; then TOUCH_EN=0. Do not reset IOX merely on one I2C error.
Start I2C at 100kHz. Pull-up estimate is 3.20k with the present 4.7k and 10k; include actual panel resistors and capacitance.
Powered-off protection: signal pins <=3.6V for VDD<1.5V; verify supply ramps and signal leakage with actual hardware.
With AON=0: R215 must pull SEL LOW even while TOUCH retains charge. GPIO hold/reset behavior is not a power guarantee.

3 LOAD-SWITCH REVIEW / MEASUREMENT LIMITS

Target for each low-voltage load rail: <0.2V within 1s after OFF. EPD high-voltage nodes are outside this target.
CT=4.7nF: TI table ~8ms at 3.3V: formula ~8.6ms (3.3V), ~11.5ms (4.4V). Actual load turn-on and inrush need measurement.
QOD=330R external plus internal RPD. Example only: 200uF, 355R, 3.3V -> 0.2V in ~0.20s with VIN still valid.
Input loss weakens QOD; timing is not guaranteed at VIN=0. Do not backfeed a rail while QOD is discharging it.
Remove measurement OR only AFTER stopping the peripheral and isolating its signals. QOD remains connected on the load side.
TOUCH: opening R408 also removes TMUX VDD; I2C pulls remain on that same isolated load rail. Verify residual voltage first.
External rail injection requires upstream OR AND QOD resistor removal, plus reviewed signal isolation; no direct supply paralleling.

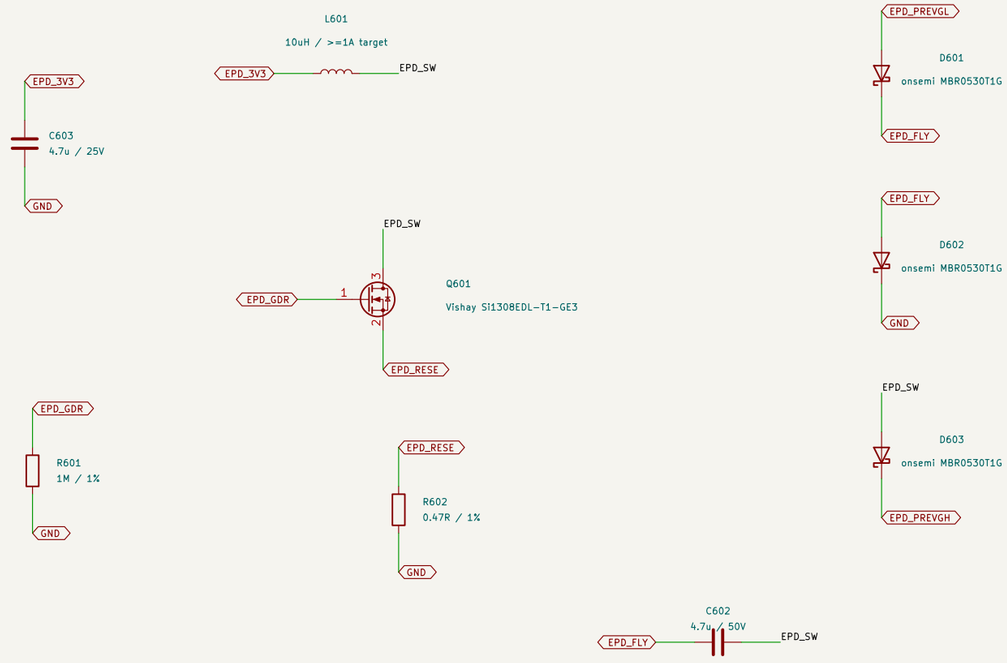
Add actual EPD-driver / SD / audio capacitance before final CT and QOD selection.
SYS -> AMP is a supply gate, not a 5V boost. No load switch provides fault-current limiting.
Refs: TI SLVSD76C pp3,13-15; SCDS390B pp3-4,20-22. Values are NOT bench-validated.

DESIGN REVIEW ONLY - NO PCB RELEASE / NO BATTERY CHARGE APPROVAL
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Sheet: /TOUCH_ISOLATION/		
File: TOUCH_ISOLATION.kicad_sch		
Title: GT911 FOUR-SIGNAL ISOLATION / DOMAIN TEST NOTES		
Size: A3	Date: 2026-11-02	**Rev: r4 DRAFT**
KiCad E.D.A. 10.0.7		Id: 7/11

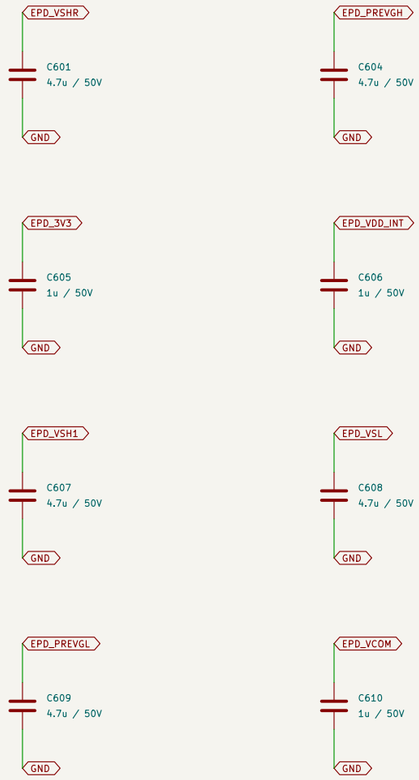
1 GOOD DISPLAY p38 BOOST TOPOLOGY / CANDIDATE PARTS

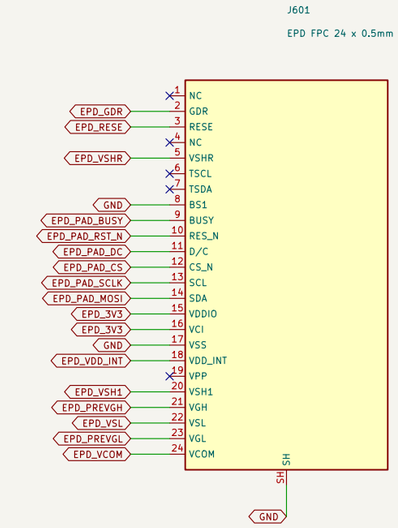
3.3V input only. Panel-generated HV: no external supply injection. Values are reference starting points, not measured.



D601: A=VGL, K=FLY; D602: A=FLY, K=GND; D603: A=SW, K=VGH.
Q601 source is RESE, not direct GND. Keep SW/FLY loops compact.
HV MLCC: X5R/X7R, effective capacitance at bias must be qualified.

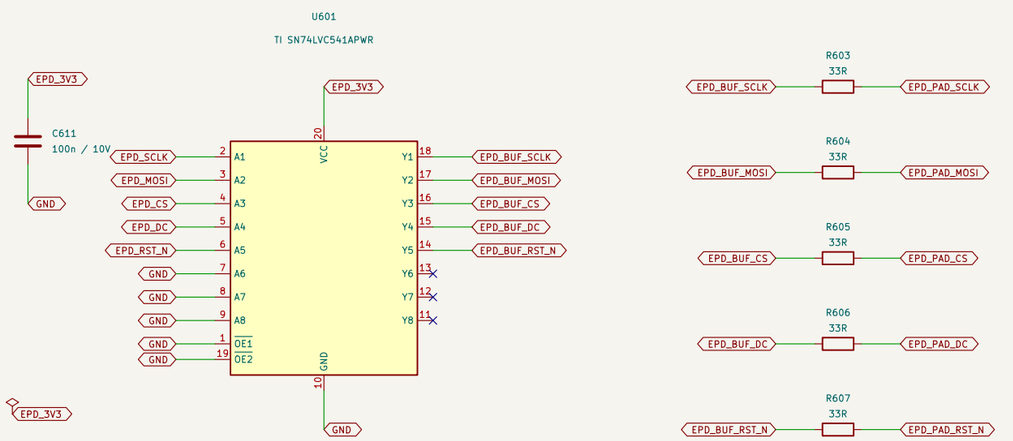
2 PANEL PINS / STORAGE AND DECOUPLING





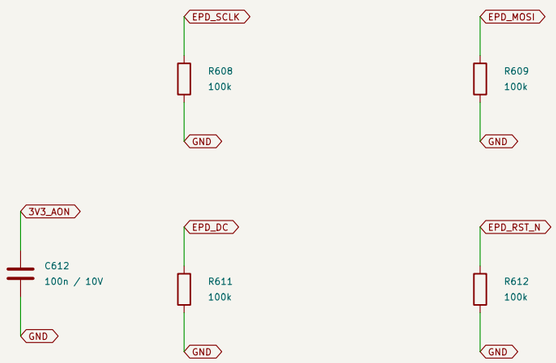
BS1=0: 4-wire SPI, write-only SDI in this board.
NC 1/4/19: unused external-temp pins6/7 left open.
VDD18 is INTERNAL: C606 only: never short to3.3V.
24 signals + SH to GND; check retained extension chain.

3 FIVE MCU OUTPUTS / EPD-SUPPLIED Ioff BUFFER

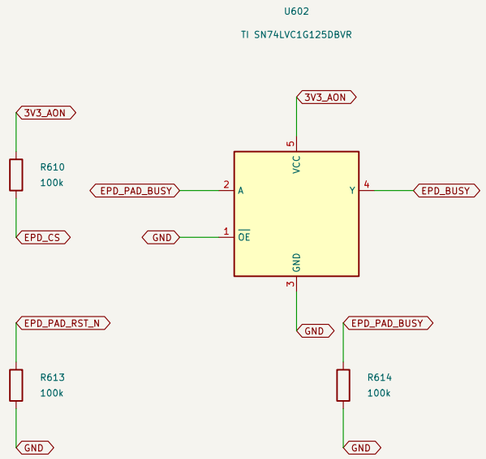


Flag: R402 populated, normal RUN only.
Unused buffer inputs tied low; outputs NC.

4 DEFAULTS / BUSY RETURN



Before ON: CS=1, RESET=0, CLK/MOSI/DC=0; wait valid rail, then reset panel.
Before OFF: finish refresh, sleep panel, hold safe signals; disable EPD supply.
Buffers do not replace reset timing or leakage/ramp testing. BUSY low while OFF is invalid.
No SDI readback. Check +/-HV, gate/current-sense and FPC before connecting the panel.



DESIGN REVIEW ONLY — NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

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Sheet: /EPD_DISPLAY/
File: EPD_DISPLAY.kicad_sch

Title: EPD REFERENCE BOOST / 24-PIN PANEL / DOMAIN BUFFERS

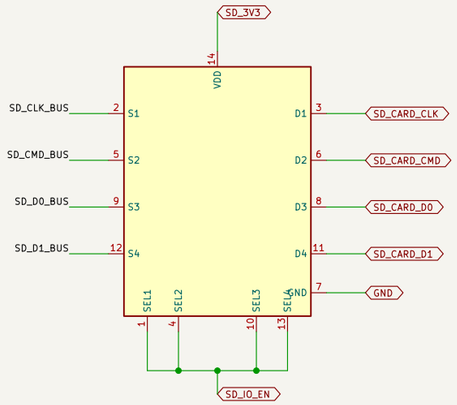
Size: A3	Date: 2026-11-02	**Rev: r4 DRAFT**
KiCad E.D.A. 10.0.7		Id: 8/11

1 SIX-LINE BUS ISOLATION / SEPARATE SD_IO_EN CONTROL

Two TMUX1511PW on SD_3V3. IOX P13 controls isolation; P01 controls card power. Six bidirectional paths; no MCU-to-card bypass.

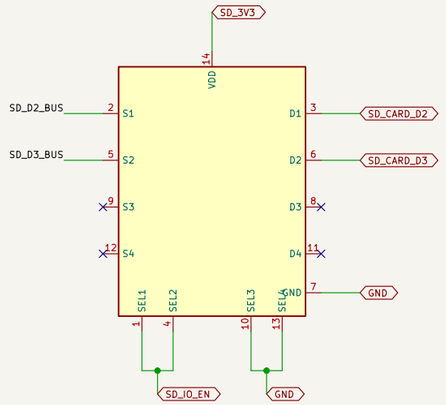
U701

TI TMUX1511PW



U706

TI TMUX1511PW



R710
22R
SD_CLK
SD_CLK_BUS

R711
0R
SD_CMD
SD_CMD_BUS

R712
0R
SD_D0
SD_D0_BUS

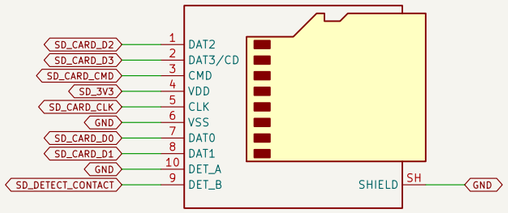
R713
0R
SD_D1
SD_D1_BUS

R714
0R
SD_D2
SD_D2_BUS

R715
0R
SD_D3
SD_D3_BUS

J701

Hirose DM3AT-SF-PEJM5



Physical card: 1 DAT2 / 2 DAT3 / 3 CMD / 4 VDD
5 CLK / 6 GND / 7 DAT0 / 8 DAT1.
No 1.8V UHS mode. Begin at 400kHz init, then20MHz target.
CD closes to GND on insertion; debounce in software.

2 CARD-SIDE PULL-UPS / DECOUPLING / DEFAULT DISCONNECT

R701
10k / 1%
SD_3V3
SD_CARD_CMD

R702
10k / 1%
SD_3V3
SD_CARD_D0

R703
10k / 1%
SD_3V3
SD_CARD_D1

R704
10k / 1%
SD_3V3
SD_CARD_D2

R705
10k / 1%
SD_3V3
SD_CARD_D3

SD_IO_EN (IOX P13): LOW isolates all six lines.
R223=10k default pull-down on control page.
Connect only after SD power has settled.

R708
100R
SD_CD_N
SD_DETECT_CONTACT

R706
100k
SD_CARD_CLK
GND

C701
100n / 10V
SD_3V3
GND

C702
10u / 10V
SD_3V3
GND

C703
100n / 10V
SD_3V3
GND

SD_3V3

PWR_FLAG assumes R405 fitted.
SD_3V3 is downstream of I_SD.

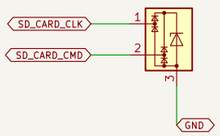
C705
100n / 10V
SD_3V3
GND

C704
10n / 50V
SD_CD_N
GND

3 CONNECTOR-SIDE ESD / NO SUPPLY CLAMP RAIL

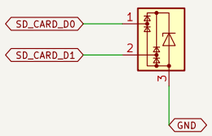
U702

TI TPD2E2U06DCKR



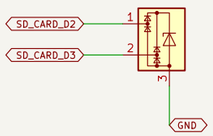
U703

TI TPD2E2U06DCKR



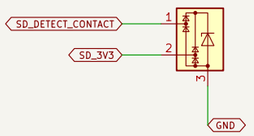
U704

TI TPD2E2U06DCKR



U705

TI TPD2E2U06DCKR



ON: SD_IO_EN=0; enable SD_EN: wait validated rail-settle time; then SD_IO_EN=1 and initialize the card.
OFF: block requests, flush/close/unmount, stop SDMMC; SD_IO_EN=0; then SD_EN=0. Never cut power on unmount timeout.
TMUX: VDD<1.5V allows signal0..3.6V; ramps/functional behavior still need validation. Spare SELs LOW; spare S/D NC.
SD power rail / GPIO hold / removal behavior / 4-bit20MHz signal integrity remain bench tests; no measured throughput claim.

DESIGN REVIEW ONLY — NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

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Sheet: /STORAGE_SD/

File: STORAGE_SD.kicad_sch

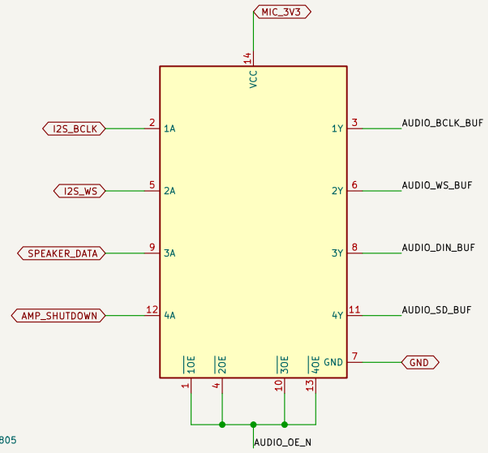
Title: MICROSD / 4-BIT BUS / INDEPENDENT SIGNAL ISOLATION

Size: A3	Date: 2026-11-02	**Rev: r4 DRAFT**
KiCad E.D.A. 10.0.7		Id: 9/11

1 FOUR MCU OUTPUTS / AUDIO_EN HARDWARE DISABLE

U801

Nexperia 74LVC125APW



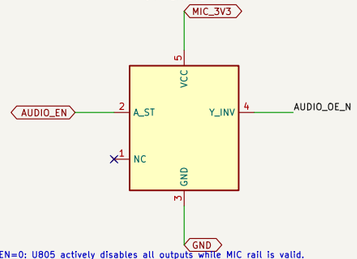
R808 22R: AUDIO_BCLK_BUF — I2S_BCLK_AUDIO

R809 22R: AUDIO_WS_BUF — I2S_WS_AUDIO

R810 22R: AUDIO_DIN_BUF — I2S_DIN_AUDIO

U805

TI SN74LVC1G14DBVR



C806 100n / 10V: MIC_3V3 — GND

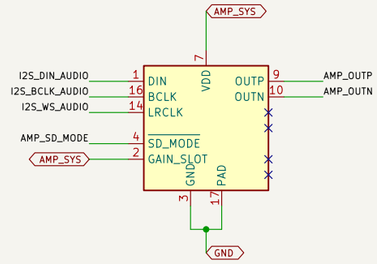
C801 100n / 10V: MIC_3V3 — GND

AUDIO_EN=0: U805 actively disables all outputs while MIC rail is valid.
At VCC=0, use the selected vendor Ioff specification; not zero leakage.
R306 already pulls MCU AMP_SHUTDOWN low; R416 pulls AUDIO_EN low.

2 MAX98357A / SYS SUPPLY / LEFT CHANNEL / 6dB

U802

MAX98357AETE+T



C802 100n / 10V: AMP_SYS — GND

R811 0R / SPK_P: AMP_OUTP — SPK_P

R812 0R / SPK_N: AMP_OUTN — SPK_N

J801

SPEAKER / BR CANDIDATE

SPK_P 1

SPK_N 2

R806 2.00k / 1%: AUDIO_SD_BUF — AMP_SD_MODE

R802 100k: I2S_BCLK_AUDIO — GND

R803 100k: I2S_WS_AUDIO — GND

R804 100k: I2S_DIN_AUDIO — GND

R805 100k: AMP_SD_MODE — GND

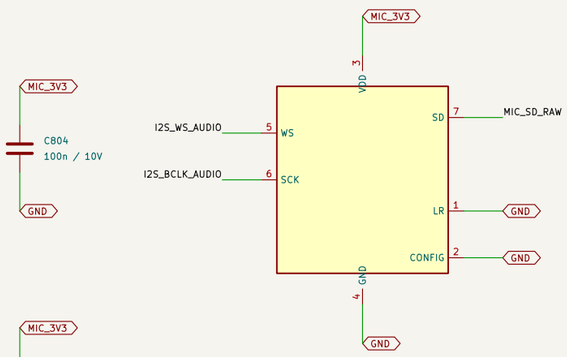
C803 10u / 10V: AMP_SYS — GND

Selected speaker: 3525 cavity / 8 ohm / 2W. Both BTL terminals float from GND.
Initial output cap 0.5W; no 5V boost. Verify low-battery THD/pop and harness.

3 PUI I2S MIC CANDIDATE / DATA RETURN ISOLATION

U803

DMM-4026-B-I2S-R

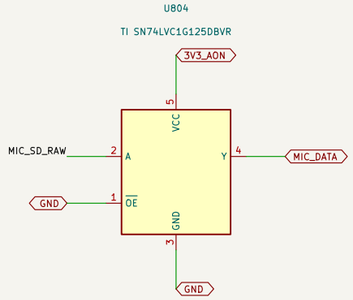


C804 100n / 10V: MIC_3V3 — GND

C805 1u / 16V: MIC_3V3 — GND

U804

TI SN74LVC1G125DBVR



R813 100k: MIC_SD_RAW — GND

C808 100n / 10V: 3V3_AON — GND

PUI pin map: 1 LR, 2 CONFIG, 3 VDD, 4 GND, 5 WS, 6 SCK, 7 SD.
Both control straps LOW. Footprint is NOT the old six-pad ICS-43434.
AON return buffer remains powered during deep sleep; budget its ICC/leakage.

4 FIRST BRING-UP CONTRACT / NOT MEASURED

Start at 48kHz, two 32-bit I2S slots, BCLK=64fs=3.072MHz.
PUI: 24-bit data / 18-bit precision; LEFT slot. 20ms is a datasheet startup reference only; discard initial frames with measured settling margin.
32/44.1/48kHz are the candidate shared set; 32k is at the MIC clock floor.
The old 16kHz firmware profile is NOT valid for this microphone candidate.
Software mixes/resamples playback into LEFT; no automatic stereo average.

ON: hold AMP_SHUTDOWN low and clocks/data low; enable AUDIO_EN;
wait for both rails; establish clocks/zeros; discard settling frames; unmute.
OFF: mute first, finish audio, stop BCLK without leaving BCLK running
with LRCLK absent; drive signals low, then AUDIO_EN=0.
MIC/AMP discharge rates differ; measure SD_MODE injection and pops.
Mute/stop before opening R411/R414. Never inject external rail power
without reviewing QOD, buffer paths and supply disconnection.

U801/U805 use the MIC rail; U804 remains on AON. Each adds leakage.
Dynamic clocks and pull-down currents belong in the active audio budget.
Actual speaker, MIC lifecycle, acoustic port and EMI remain review gates.

MIC_3V3

AMP_SYS

Flags: R411/R414 fitted.
Not a power-fault model.

DESIGN REVIEW ONLY — NO PCB RELEASE / NO BATTERY CHARGE APPROVAL

ESPaperPlay / xiaobaoaaaaaa

Sheet: /AUDIO/
File: AUDIO.kicad_sch

Title: AUDIO / I2S INPUT ISOLATION / MONO PLAYBACK + MIC

Size: A3	Date: 2026-11-02	**Rev: r4 DRAFT**
KiCad E.D.A. 10.0.7		Id: 10/11

CHARGE TEMPERATURE WINDOW / r5 DESIGN ASSUMPTION

Pack NTC assumed Honeywell 175-103LAE-301 Rev.B. Actual ZON.CELL sensor identity remains unverified.

VIN_5V is the same selected input as BQ24074. Temperature faults pull TS low; safety timer pauses, not resets.

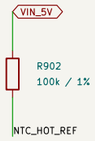
REAL PACK SENSOR (J103 on page 2)

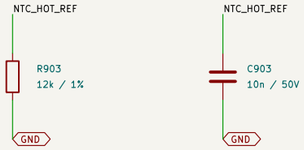




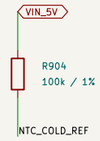
J103.1 = BAT_NTC
J103.2 = GND
R25 assumed 10k, +/-3%
Open -> cold; short -> hot.

HOT REFERENCE



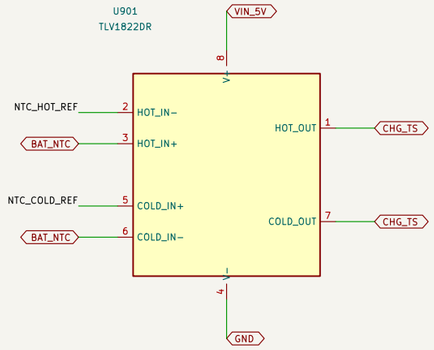


COLD REFERENCE





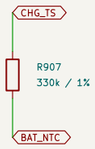
DUAL OPEN-DRAIN WINDOW





HOT: BAT_NTC < HOT_REF
COLD: BAT_NTC > COLD_REF
Either fault sinks CHG_TS.
No added TS output capacitor.

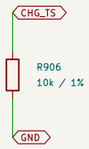
SHARED POSITIVE FEEDBACK / HYSTERESIS





5V nominal model: cold pause 14.8C; hot pause 40.1C. Resume example: 15.5C / 39.5C.
Resume example assumes fault TS=0.1V. Model budgets / corners: review-r5/temperature.
R108 remains DNP; J105 open. Maintenance CE enable does not bypass this TS window.

BQ24074 TS INTERFACE



LAYOUT / TEST: keep BAT_NTC, reference and TS traces away from switch nodes; place U901 bypass locally.
Verify power insertion/removal, resistor substitution, actual pack R-T, threshold chatter and charger thermal behavior.
POR/deglitch figures are not a guaranteed startup timing proof. No independent single-component fault guarantee.

DESIGN REVIEW ONLY - NO PCB RELEASE / NO BATTERY CHARGE APPROVAL
ESPaperPlay / xiaobaoaaaaaa
Sheet: /CHARGE_TEMP/
File: CHARGE_TEMP.kicad_sch

Title: Battery temperature window / TS pause

Size: A3	Date: 2026-11-04	**Rev: r5 DRAFT**
KiCad E.D.A. 10.0.7		Id: 11/11